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Unintended consequences of emerging communication technologies: Instant Messaging in the workplace

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Abstract

With increased global connectivity, managers are faced with new technologies and rapid organizational changes. For instance, organizations may adopt emerging technologies such as Instant Messaging in order to increase collaboration at a distance and to decrease communications costs. However, the impact and implications of these technologies for managers and employees often go far beyond the original intent of the technology designers. Consequently, in this study, instant messaging (IM) and its use in organizations were investigated through interviews with employees. Results suggest that critical mass represents an important factor for IM success in the workplace that IM symbolizes informality, and that IM is perceived to be much less rich than face-to-face communication. Further, results demonstrate that employees use IM not only as a replacement for other communication media but as an additional method for reaching others. With IM, employees engage in polychronic communication, view IM as privacy enhancing, and see its interruptive nature as unfair. The paper concludes by discussing research and practice implications for organizational psychologists.

Keywords: Computer-mediated communication; Electronic monitoring; Media richness; Polychronic communication; Privacy; Fairness

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1. Introduction

Managers and employees in today's business environments are approaching a technological communication frontier, with a panorama of technologies at their disposal. The workhorses of today's organizations are communication media such as email, teleconferencing, videoconferencing, and most recently, Instant Messaging. Market analysts have predicted that corporate Instant Messaging will grow from 18.3 million in 2001 to 229 million users worldwide in 2005 (Mingail, 2001). This sudden and unplanned emergence of corporate Instant Messaging gives rise to many questions regarding its effective use to support organizational communications.

Instant Messaging (IM) represents a communication technology that allows employees to send and receive short text-based messages in real-time and to see who else is 'online' and currently available to receive messages. Based on the 1988 Finnish technology Internet Relay Chat, IM was originally intended to allow home internet users to converse with family and friends (Goldsborough, 2001). However, as managers began to recognize this tool's potential to support informal interactions (Nardi, Whittaker, & Bradner, 2000; Pauleen & Yoong, 2001; Perry, O'Hara, Sellen, Brown, & Harper, 2001), more and more corporations began installing IM software on employees' workstations. This initial use has since spread outside the organization to communications with customers (Guan & Alkinkemer, 2002).

IM systems differ in two main ways from the chat systems on which they were based. First, IM messages generally arrive singly, while multiple conversations often appear in a scrolling window of the computer screen with chat systems (Segerstad & Ljungstrand, 2002). Second, IM systems contain a presence awareness capability. Presence awareness, also called peripheral awareness (Simone & Bandini, 1997), distributed awareness (Dourish & Bly, 1992), background awareness (Lee, Girgensohn, & Schlueter, 1997), and awareness moments (Nardi et al., 2000), involves having a general sense of who is around and what they are doing (Dourish & Bly, 1992; Zweig & Webster, 2002). Unlike performance monitoring, presence awareness represents a type of peer monitoring designed to enhance communication between colleagues (Zweig & Webster, 2003). With IM, this capability allows users to see status indicators representing other users who are 'online' and currently available to receive messages as well as when they were last actively using the system. Further, the awareness function of some IM products is able to automatically determine when an employee has logged on or logged off his or her computer as well as the amount of time the employee's IM system has remained idle. Some IM systems also allow the employee to turn off the status indicator altogether, denying others access to their presence information.

Although some research has examined chat systems (e.g., Dietz-Uhler & Bishop-Clark, 2001; Orvis, Wisher, Bonk, & Olson, 2002), few studies have examined IM systems in depth (Olson & Olson, 2003) and less is known about how they are actually used in practice (Isaacs, Kamm, Schiano, Walendowski, & Whittaker, 2002). Many of the existing IM studies demonstrate the feasibility of prototype systems rather than examine how IM systems are used in the workplace (Grinter & Palen, 2002). As Handel and Herbsleb (2002) indicate, "important studies of instant messaging have looked primarily at use by research groups that developed the tools and it is not clear how typical this use is in a general workplace population''. In contrast to previous IM research, we draw on multiple theories to help explain IM use in the workplace and to suggest implications for organizational psychologists. Further, by studying IM, we are responding to calls for more research on monitoring systems that do not focus on performance (e.g., Stanton & Weiss, 2000). To accomplish this, we explore two research questions:

1. Why does an employee choose to use IM rather than another medium when communicating?

2. How are IM systems used in the workplace?

In the following section, IM systems are described and existing research is outlined. Then, theories around communication and electronic monitoring are explored as they might apply to IM. Next, the research method employed and the results of this research are described. Finally, the implications of these results for managers and employees are discussed, future areas of research are proposed, and several strengths and limitations of the study are outlined.

2. Instant Messaging research

Instant Messaging has been described as a technology that allows ''users to set up a list of partners who will be able to receive notes that pop up on their screens the moment one of them writes and hits the send button'' (Castelluccio, 1999, p. 35). Many IM systems can be downloaded for minimal cost from the internet or come free with other software (such as ICQ Chat, AOL Instant Messenger, MSN Messenger and Yahoo Messenger). Further, several IM systems have been developed specifically for corporate use to enhance real-time interactions between co-workers (e.g. Lotus SameTime, Jabber Instant Messenger, and Bantu Instant Messenger) and more are under development (Pruitt, 2003).

Given that IM, email, and chat all send electronic messages, IM is often viewed as similar to these other media. However, there are significant differences. For example, unlike email users, most IM users can only send messages to others using the same IM system, IM users send and receive messages in real time (much faster than traditional email), and IM messages are usually shorter than email messages. One difference that sets IM apart from both email and chat is IM's presence awareness functionality. Further, unlike chat, IM systems are used to communicate one-on-one and with small groups of known others (Grinter & Palen, 2002). Thus, IM systems are viewed as more private but also more intrusive than chat (Handel & Herbsleb, 2002).

Table 1 summarizes existing IM studies, most of which were presentations at human–computer interaction (HCI) conferences. Because IM is an emerging technology, many of these studies were conducted by the developers of the IM systems using colleagues in their own organizations (e.g., Isaacs et al., 2002) or investigated teenagers and university students (e.g., Grinter & Palen, 2002). Further, many of these studies did not draw from a theoretical base. There are some notable

Table 1
Empirical research on Instant Messaging[a]

Source	Theory base	IM system	Participants	Methods	Findings
Bradner and Mark (2002)	Social impact and social identity	AOL	98 Undergraduate students	Experiment	No differences between IM and videoconferencing in cooperation, persuasion, or deception
Cutrell, Czerwinski, and Horvitz (2001)	Memory, timing, task type and switching	Simulated IM (similar to Microsoft Messenger)	16 Participants between the ages of 20 and 57	Experiment	Participants who received an IM were slower overall, needed more reminders, and were interrupted more at the beginning than later in their task
Grinter and Palen (2002)	–	Popular IM systems (AOL, ICQ, MSN, or Yahoo!)	16 Teenagers	Interviews	Participants reported being involved in multiple, concurrent IM conversations and in engaging in another computer-based task at the same time
Hansen and Damm (2002)	–	Yahoo! (with a researcher-developed prototype called "Knight")	7 Software developers in researchers' workgroup	Informal participant observations, interviews, and logs of transcripts	Developers shifted between IM and other media (e.g., telephone) when required; status (awareness) information useful only if people kept this information up-to-date
Herbsleb, Atkins, Boyer, Handel, and Finholt (2002)	Critical mass	A researcher-developed prototype IM tool ("Rear View Mirror")	6 Distributed teams in researchers' organization	Semi-structured interviews and focus groups	Difficult to introduce IM tool into the workplace: 90% of possible users dropped off
Isaacs et al. (2002)	–	A researcher-developed prototype IM tool ("Hubbub")	32 employees in researchers' organization	IM logs	Each participant moved out of the message window an average of 3.7 times/conversation; participants switched to another communication medium 17.8% of the time

Isaacs, Walendowski, Whittaker, Schiano, and Kamm (2002)	–	A researcher-developed prototype IM tool (“Hubbub”)	28 employees in researchers’ organization; some data from 437 users (half unknown)	IM logs	In 85.7% of conversations, at least one person multi-tasked (generally on unrelated activities)
Nardi and colleagues (2000)	Task fit	[not indicated]	20 Employees from two IT companies and an independent contractor	Ethnography	IM used for ‘interaction’ (information exchange) & ‘outeraction’ (reaching out to others); IM often monitored while using other media
Pauleen and Yoong (2001)	–	ICQ	7 Professionals in a virtual team training program	Grounded action research	IM used to support informal conversations in virtual teams
Schiano, Chen, Ginsberg, and Gretarsdottir (2002)	–	[not indicated]	26 Teenagers	Ethnography	IM was used for immediate, casual contact – young girls used it most; although some teens IM’ed with up to 5 others at once, most did so with an average of 1–2 others
Segerstad and Ljungstrand (2002)	–	University system called “WebWho”	University students	IM logs over 16 months	Humour highest topic category; task-related second-highest category
Voida, Newstetter, and Mynatt (2002)	Socio-linguistics	[not indicated]	8 Members of a university research lab	Interviews, observations, IM logs	Users multitasked

[a] We include research that examines IM technologies (that is, systems with both individually addressed synchronous text messaging and a presence awareness component) but not chat systems. Chat systems tend to be more public and may not have a presence awareness component. Therefore, we excluded studies such as Handel and Herbsleb (2002) because only results for group chat were reported (although IM was part of their system).

exceptions. For instance, Nardi et al. (2000) completed an ethnography, interviewing and observing 20 employees using a corporate IM system. They built on communication media theories to demonstrate that IM users exhibited the expected signs of 'interaction' (information exchange) as well as the less anticipated signs of 'outeraction' (communicative processes used to reach out to others) when using the system.

What this emerging literature demonstrates is that IM users often multi-task, or use IM with other media. Using multiple media at the same time has been called simultaneity in the HCI literature and has been defined as "the superimposition of activities on top of one another, so that multiple activities can be performed at the same time" (Perry et al., 2001, p. 343). For example, an individual driving a car and talking on a cell phone is exhibiting simultaneity. In contrast to the HCI literature, other literatures use the term polychronicity for the concurrent completion of multiple tasks (Bluedorn, 2000); we use this term in the remainder of the paper.

Although little theoretically informed research on IM in the workplace exists, researchers should not be at a loss to explain use of this particular technology. Existing literature and theories may be useful in investigating the use of IM in the workplace. Instant messaging is a tool that enables communication and, as such, traditional theories that explain the use of communication media, such as critical mass, symbolic cues, and media richness, may increase our understanding of IM systems. In addition, newer theories of polychronic communication should help to inform IM use. Further, IM systems employ presence awareness technologies which represent a type of monitoring; thus electronic monitoring theories may also help to inform IM system usage.

3. Theoretical background

We propose that traditional communication theories (such as critical mass), emerging communication theories (polychronic communication), and theories around electronic monitoring technologies (such as privacy and fairness) will help to explain the use of IM in the workplace.

3.1. Communication theories

Critical mass. The defining feature of all communication technologies is that they "require multiple users and cannot be used successfully by one person acting alone" (Soe & Markus, 1993, p. 213). Thus, the success of a communication technology is not only reliant on an individual's use of the technology, but on others' responses to this use (i.e., do they answer, acknowledge, or reply to the message sent?). The theory of critical mass states that "once a certain number or proportion of users (critical mass) have been attracted, use should spread rapidly throughout the community" (Markus, 1987, p. 500). One reason offered for this rapid diffusion is that using a technology that has reached critical mass allows users to communicate to the largest number of people with the least effort. In the absence of critical mass, use of the technology may decrease and even cease entirely. Thus, for corporate IM, critical mass theory would suggest that employees will be more likely to use the IM technology if intended message recipients are also using it on a routine basis.

Symbolic cues. Based on the theory of symbolic interactionism, it has been suggested that "the symbolic cues conveyed by the medium itself above and beyond the literal message" can affect media choice (Trevino, Daft, & Lengel, 1990, p. 74). In a sense, the selected medium becomes a part of the message, revealing the intent of the sender (Fulk, 1993; Trevino et al., 1990; Trevino, Lengel, & Daft, 1987; Webster, 1998; Webster & Martocchio, 1995). As such, certain media are more appropriate for demonstrating legitimacy, teamwork, formality, informality, urgency, or caring. For example, a face-to-face meeting might convey caring while an official memo might convey legitimacy and formality. Communication via IM may suggest a light and informal tone, non-authoritative conversation, and the breaking down of hierarchical barriers. The symbolic cues that a particular medium conveys are based on the subjective norms of the situation and may differ across organizations or across workgroups. Thus, IM may be used to communicate a sense of urgency in one workgroup and a lack of formality in another workgroup.

Media richness. Fulk suggested that "[c]ommunication technologies in particular link disparate entities in a seamless web that engages joint sensemaking in the process of mediated interaction" (1993, p. 922). Thus, the capability of a communication technology to enable this 'joint sensemaking' or to facilitate shared meaning is extremely important. Trevino and colleagues (1987) described this capability in terms of the 'richness' of the media. Rich media are those that provide instant feedback, allow verbal and non-verbal cues, use natural language, and convey emotion. Rich media are thought to be best when communicating ambiguous ideas or concepts. Face-to-face communication would rate high on richness and email-based communication would be considered lower in richness. It has been suggested that IM is a combination of face-to-face and email (Segerstad & Ljungstrand, 2002) and would fall somewhere between the two on the media richness scale. For instance, Segerstad and Ljungstrand found that while the IM system was used by the students "without any teacher interference or control, for whatever purposes the students might see fit" (2002, p. 149), task-related topics were the second highest message content category, accounting for 15.8% of the total messages sent during a 16-month period. This IM system was used for more than just social processes and relationship building. Similarly, if IM users in a corporate environment think that IM systems enable rich communication, they may be more likely to use IM more often in a variety of situations.

Polychronic communication. Although researchers have been studying the notion of polychronicity since it was first introduced by anthropologist E.T. Hall (Hall, 1959), research on polychronic communication is just beginning. Polychronic communication represents "the managing of multiple conversations at once within a given time period" (Turner & Tinsley, 2002, p. 4). However, it has been observed that the user is not necessarily "substituting one activity for another, or dealing with a task more quickly" (Jaureguiberry, 2000 as cited in Perry et al., 2001,

p. 343). Past research has demonstrated that polychronicity can relate to important employee outcomes such as role overload and the overlap of work and non-work (e.g., Benabou, 1999; Kaufman, Lane, & Lindquist, 1991). As noted earlier, past research on IM has found that a substantial amount of polychronic communication takes place.

3.2. Electronic monitoring theories

Privacy. The presence awareness capabilities of IM allow employees to know which colleagues are currently present and available for communicating. Previous research into video awareness systems has found that some users are reluctant to accept these systems (e.g., Lee et al., 1997; Whittaker, 1995; Zweig & Webster, 2002). Users are concerned that these video tools are an invasion of privacy and could be used for monitoring purposes. IM users may likewise believe that such systems give away too much information about their whereabouts and activities (Godefroid, Herbsleb, Jagadeesan, & Li, 2000). Consequently, to counteract privacy concerns with IM, some teenagers have blocked certain friends in order to hide from them or have used multiple screen names or aliases (Grinter & Palen, 2002). Similarly, university lab members have maintained separate user names for members of different social clusters (Voida et al., 2002). However, unlike video awareness, IM senders do not know for certain that the IM recipients are in their workspaces; this "plausable deniability" of presence provides IM users with greater privacy than video presence users (Nardi et al., 2000).

Although IM may increase concerns about privacy because of its presence component, it may also be seen as privacy-enhancing with respect to local others. For instance, studies of teenagers have demonstrated that IM may give them more privacy when at home. This is because IM "is a quiet technology that is easily integrated into the conduct of other activities... Use can be unobtrusive, go unnoticed, or even be covert... teenagers use IM to carve out a private world within the public space of the home" (Grinter & Palen, 2002, pp. 26–27). Thus, IM users may find the presence component an invasion of privacy but the opportunity to quietly communicate with others an enhancement of privacy.

Fairness. Past research has demonstrated that employees view electronic monitoring systems not only as an invasion of privacy but as unfair (e.g., Alge, 2001) and it has been well-established that fairness plays a key role in determining reactions to organizational processes (Ambrose & Adler, 2000). For instance, video awareness systems allow others to know that employees are at their workstations: this knowledge may be viewed as unfair because employees have less control over their own work as others interrupt them (Zweig & Webster, 2002). Similarly Herbsleb et al. (2002, p. 171) suggested that

> there are likely to be some individuals whose attention is in more demand than others. IM may impose an undue burden on them, but mostly provide benefits for others, i.e., those who want to reach them. While other forms of CMC [computer-mediated communication], such as e-mail, also have this potential, the synchronous nature of IM makes it harder to ignore.

Some employees view IM "as further encroaching on their time to do 'real' work" (Herbsleb et al., 2002, p. 176). Even the notification of an incoming message can negatively affect task performance (Cutrell et al., 2001). Thus, we expect that employees may also perceive IM as interruptive. However, Nardi et al. (2000) found that users view IM as less interruptive than the telephone or face-to-face communication because IM allows employees to "negotiate availability" by consulting presence information.

4. Method

Instant Messaging and its use in organizations were investigated using the case-study approach. A snowball sampling method was used to find 19 employees from four separate organizations, or cases, to take part in this research. The position (management vs staff) and gender of the participants are shown in Table 2. For all cases, the IM systems were internally distributed and/or sanctioned by the organization (i.e., not downloaded from the Internet for personal use by individuals).

The first author, who has qualitative research methods training, conducted the interviews and used a tape-recorder as well as note-taking to capture the information. She spoke face-to-face with interviewees from organizations A and B and by telephone with interviewees from organizations C and D. Semi-structured interviews were conducted one-on-one with each participant and were approximately 45 minutes in length. A set of general questions were used to guide the interview and to determine the individual's technology use patterns and practices, frequency of use, experienced positive outcomes, experienced negative outcomes, and general perceptions about IM in a corporate environment. For organizations A and B where the interviews were conducted face-to-face, the researcher received a brief tour of the working areas and observed at least two participants from each organization interacting with the IM technologies in their natural working environments. At the end of the interview, participants were given a one-page questionnaire, adapted from Trevino, Webster, and Stein (2000), which attempted to determine the symbolic cues associated with the IM medium. [1] Respondents were asked to indicate (using a five-point Likert-scale) to what degree sending a message via Instant Messaging signified each of 28 symbolic cues.

Transcriptions of each interview and researcher notes were loaded into qualitative analytic software (i.e. NU*DIST 6) for coding and analysis. Using the software, the transcriptions were coded by the first author according to the various theoretical perspectives covered above. Additionally, transcripts and research notes were coded according to the position (managerial vs employee), gender, and organizational affiliation of the interviewee. A second individual, blind to the research questions, coded a random sample (25%) of the interview transcripts in an attempt to minimize

[1] The complete 28 items are available from the second author.

Table 2
Case descriptions

	Case A	Case B	Case C	Case D
Type of Org.	Privately owned, high-growth technology firm	Publicly traded, providing technical solutions for private and public businesses	Publicly traded telecommunications firm	Privately co-owned, developer of technical HR tools and websites
Founding	1995	1970s	1990s	1994
Size	130 Employees	2400 Employees	30,000 + employees	2
Location	Two main offices in one region North America	Five countries, headquartered in the US	150 Countries worldwide	Two individuals located at opposite ends of a North American city
Management style	More autocratic, centralized	More laissez-faire, decentralized	Encourages innovation but still quite structured	Not a typical 9-5 organization. Flexible
# of interviewees[a]	7 Interviewees (3M managers, 2F managers 1F admin assist., 1M employee)	8 interviewees (3M managers, 5M employees)	2 interviewees (1M managers, 1M employee)	2 interviewees (2M co-owners)
Type of IM tool used	In-house developed. Available on/off-site. No archiving of messages. Availability updated manually.	Lotus SameTime Connect. Available on-site only. Archiving only if users manually save messages. Availability updated automatically.	AOL Instant Messenger. Can communicate with those both internal and external to the company. Availability manually updated.	ICQ. Communicate with each other and sub-contractors. All conversations archived. Availability automatically updated
Length of time IM used	>5 years	6 months	>3 years	4 years
IM use discretionary?	No (because of set rules)	Yes	No (because of expectations)	Yes
Degree of dispersion of Department Members	Low: Teams usually at one location.	Medium: Teams usually spread over multiple locations with >1 individual at each location.	High: Teams usually completely spread out across countries and between home and office.	Medium: Individuals are in same city but rarely meet face-to-face

[a] M = Male and F = Female. So, 3M managers means three male managers.

idiosyncratic coding (Miles & Huberman, 1994). The results of the first and second coders were compared and resulted in an inter-rater reliability of 80%.

5. Results

The characteristics of the four companies studied are summarized in Table 2. The sample contains both large and small organizations, centralized and decentralized management styles, and varying lengths of time IM had been used in the company. Some of the IM tools employed were true corporate IM systems (e.g., In-house developed system and Lotus SameTime Connect) while others were mainstream IM systems applied to the corporate environment (e.g., AOL and ICQ).

5.1. Why does an individual choose to use IM?

In order to answer the first research question, "Why does an employee choose to use IM rather than another medium when communicating?" interviewees were explicitly asked to explain, for their most recent IM use, why they used IM instead of another communication medium. In addition, answers to this question emerged during other sections of the interviews. The explanations that surfaced most frequently reflect three main themes: critical mass, symbolic cues, and media richness.

Critical mass. Critical mass may be one reason an employee uses IM to communicate instead of another medium. Considering that all employees at Company A were required to use the IM tool and less than 20 employees at Company B used the tool, the data in Table 3 become particularly significant. Perhaps the critical mass of users has not yet been reached at Company B. One interviewee noted, "If it becomes widely deployed in our company and more people get on it I would say that the usage of it would, for myself, go up". Evidently, the IM system at this organization does not yet allow users to communicate with the largest number of people with the least effort.

Symbolic cues. Analysis of interview data revealed that 13 (68%) of interviewees felt that sending a message through IM suggests an informal tone. One interviewee suggested "telephone's too formal... you've got to be nice and you've got to say all the niceties... with instant messaging... you can just get it done". Analysis of the symbolic cues questionnaire also demonstrated that users choose to use IM to indicate an informal tone; they also use IM when the intent of the message is to get

Table 3
Critical mass

	Mentioned critical mass	Did not mention critical mass
Case A	0	7
Case B	7	1
Case C	0	2
Case D	1	1

attention, to get a quick response, and to symbolize efficiency. Further, they felt that messages sent through this medium are less appropriate for conveying formality, symbolizing official communication, or for emotional messages.

Media richness. Overall, interviewees did not think that IM is a very rich medium. Only one interviewee felt that IM was a good medium for conveying meaning and emotion. Only three interviewees reported that IM could be used to effectively communicate ambiguous ideas and concepts. Quick responses/instant feedback was the only dimension of media richness on which IM systems rated highly (mentioned by 10 individuals or 53% of total interviewees). For example, one interviewee stated that she used IM to communicate because "it was more immediate and would have a faster flow of conversation". Thus, IM seems to be considered *rich* only when measured on one dimension of the media richness concept.

5.2. How are IM systems used?

In order to answer the second research question, "How are IM systems used in the workplace?", interviewees were asked to describe various situations in which they communicated using IM. Four general themes emerged: IM as a replacement technology, polychronic use of IM, IM to enhance privacy, and IM as an unfair technology.

Replacement technology. As demonstrated in Table 4, IM was often used as a replacement technology, substituting for other methods of communication such as telephone, email, or face-to-face meetings. However, six of the interviewees reported that if the IM tool had not been available, no communication would have taken place: these employees viewed IM as an additional method for reaching others.

Polychronicity. IM use often occurred when one or both of the conversants were using IM at the same time as another communication medium. Three types of polychronicity emerged. First, an individual might communicate with another using two media. For example, John may use the telephone to communicate verbally with Jane and use IM to quickly send her URLs to websites. This type of polychronicity can be quite useful for sending multiple forms of communication (e.g. verbal and text) within a single conversation with an employee.

The second type of polychronicity occurred when an employee communicated with one person using one medium and initiated a conversation with a second person

Table 4
IM as a replacement or addition to other media[a]

	Replacing telephone	Replacing email	Replacing F-T-F	Addition
Case A	4	3	4	2
Case B	4	5	1	3
Case C	2	1	0	0
Case D	2	0	0	1
Total	12	9	5	6

[a] Some interviewees gave more than one response.

using a second medium. An example of this type of polychronicity would be John talking on the telephone with Jane while engaging in an IM conversation with Mary. As one interviewee described it "Sometimes [the instant message] refers to the conversation you're having on the phone ... you sort of message the other person to verify your facts."

The third example of polychronicity is a special type that we will call *queue jumping*. This occurs when an individual communicates with one person using one medium while a second person attempts to initiate a conversation with him or her through an alternate medium. In our example, John would be talking with Jane in his office and others may be waiting outside to speak with him. Mary knows that John is already busy and attempts to use IM to jump ahead in the queue and get his attention. As one user described it: "So usually what I do with the managers is I get up to go see if they are busy and I'll peek in unobtrusively [and if] they've got someone there, that's when I use [IM]."

Twelve of the interviewees reported that they used IM for at least one of the forms of polychronicity. However, there can be disadvantages to polychronic communication. For instance, one interviewee suggested that trying to concentrate on too many tasks at once could result in the deterioration of one or both of the conversations.

Privacy. The argument that the presence awareness component of IM is an invasion of privacy was not supported as 15 (79%) of the interviewees stated that privacy was not a concern when using IM. In fact, it was discovered that IM systems could actually protect and maintain privacy (consistent with research on the privacy-enhancing nature of IM around teenagers' families). In open-office concept environments (such as cubicle farms or shared offices) both face-to-face and telephone conversations are easily overhead by coworkers. IM was often the only way employees in this environment felt they could converse with other employees and managers in a private way. As one interviewee noted, "It's more private for me to have an instant message because there's no talking involved and the people around me can't hear".

Fairness. An issue relating to fairness, specifically the interruptive nature of IM tools, emerged during the interviews. Although no direct questions regarding interruptions were asked, over half the interviewees noted that IM technology has the ability to interrupt work. Interviewees reported five different ways that IM could be interruptive. The most significant concern was the tendency for IM messages to break one's concentration while focused on another task. One interviewee remarked, "I think this ICQ stuff will [be] another challenge to people being focused at work". However, some interviewees believed that while the technology is interruptive, it is a less interruptive medium than telephone or face-to-face contact.

6. Discussion

Results suggest that critical mass represents an important factor for instant messaging success in the workplace, that IM symbolizes informality, and that IM is perceived to be much less rich than face-to-face communication. Further, results demonstrate that employees use IM not only as a replacement for other communication media but as an additional method for reaching others. Employees engage in polychronic communication with IM, taking part in multiple forms of multitasking. For instance, some use IM to "jump the communication queue" in front of waiting others. Finally, employees view IM as a private medium for communication but see its interruptive nature as unfair. These findings have important implications for research and practice.

6.1. Implications for research and practice

The implications of emerging technologies for managers and employees often go far beyond the original intent of the technology designers. For example, designers are developing new IM systems with, for example, short tunes to identify when buddies become active, visible cues indicating colleagues' level and duration of activities, indications when colleagues are typing in their IM windows, and both text and musical IM messages (e.g., Isaacs et al., 2002). While these emerging IM technologies may increase connections, they will also result in increased interruptions to the work of others, with the potential for decreased performance. These as well as other implications for researchers and practitioners are outlined next.

Like more traditional communication media, theories of critical mass, symbolic cues, and media richness play a role in explaining IM use. For instance, the finding that critical mass is important for IM success in the workplace has implications for the deployment and implementation of IM systems. One traditional technique of system implementation involves the use of an initial pilot of the system, or a full-scale version of the system installed at one site or in one business unit "so as to work through any problems before the system is implemented in all the others" (Kendall & Kendall, 2002, p. 206). The use of such a pilot method may not allow the full benefits of the IM tool to emerge. The theory of critical mass suggests that users who are given a pilot version of the software may not use the tool to its full potential since any business contacts outside their department or business unit are not yet on the system. Even after IM is fully deployed across the company, the behavior and habits of those early users may be difficult to change. If the pilot method is used to determine the potential of IM systems in a particular business environment, organizational implementers and trainers must be aware that they might be limiting the full benefits that can arise from using an IM system.

Many interviewees suggested that using IM to send a message usually symbolizes the informality of the message. Further, some stated that IM should not be used for formal communications because not all IM systems automatically save instant messaging conversations. Keeping such a 'paper trail' can by very important in today's sensitive business environment. Thus, those who deploy IM systems should train employees to keep important and formal communications (such as giving orders and assigning responsibility for tasks) to more traceable formats such as emails and paper documents.

This study goes beyond traditional communication theories to propose that the concept of polychronicity is key to understanding the use of technologies such as IM. The results of this study suggest that IM will likely be used simultaneously with other communication media or even that the communication itself may be performed simultaneously with other tasks. However, completing multiple communication tasks at the same time is not necessarily always a 'good' thing. Researchers should investigate the effect of polychronic communication using IM on quantitative performance, qualitative performance, and working relationships. Organizational trainers should keep the potential for these polychronic behaviors in mind when introducing employees to IM systems.

Unlike past research on video presence awareness systems revealing significant privacy concerns (e.g., Zweig & Webster, 2002), the results of this investigation suggest that IM may actually protect and maintain employees' sense of privacy. This finding has important implications for organizations considering voice-based IM systems (e.g., Anonymous, 2002). While voice may have its benefits, the privacy-enabling properties of text-based IM make it extremely valuable in a corporate environment. Thus, organizations should ensure that text remains a component of any IM system adopted.

Consistent with research on video presence systems, fairness issues did emerge as an important issue. Therefore, the problem of IM interrupting employee work patterns should be investigated in more depth. From the message initiator's point of view, IM may be seen as a useful "electronic foot-in-the-door" (Gueguen, 2002), whereas from the message recipient's point of view, IM may be seen as an annoyance. Thus, researchers as well as organizational trainers should be aware of the potential advantages and disadvantages of the interruptive nature of IM tools.

While this research has important practical implications for instant messaging in particular, it also has implications for collaborative technologies more generally. Integrated systems are being created to provide not only instant messaging and presence awareness but shared whiteboards, audio and video conferencing, and collaborative document authoring and management (e.g., Qu & Nejdl, 2001). Our research suggests that not all of these features will be perceived positively by employees, nor will they all have positive impacts on performance. Thus, organizations should consider the possible positive as well as negative impacts of IM and collaborative technologies.

6.2. Strengths and limitations

This study has several strengths that allow it to make a meaningful contribution. It represents one of the first theory-based investigations of IM in the workplace. This research also responds to calls for more in-depth research on Instant Messaging (e.g., Olson & Olson, 2003) and, more generally, to calls for further investigations into the interruptive nature of technologies (Davis, 2002; Zweig & Webster, 2002) and electronic monitoring in the workplace (Stanton & Weiss, 2000). Additionally, this study extends past research on communication media to include polychronicity, privacy, and fairness.

There are two main limitations to this research. First, case study inquiry should rely on "multiple sources of evidence" (Yin, 1994, p. 12). Although the researchers attempted to collect data from multiple sources (e.g. observation of employees' IM use, observations of interviewee reactions, demonstrations of the IM products, and interviews with management), almost all of the data were gathered from semi-structured interviews with employees. Second, the particular sample used in this research was a sample of convenience. All interviewees came from companies in the high-tech industry and no non-users were part of the sample. Thus, the interviewee group may be more enthusiastic than the average employee about technologies such as IM. Results from this sample must be generalized with caution to the general IM employee population.

7. Conclusion

The quest for instant connections and enhanced communication has led to the adoption of a multitude of technologies designed to speed up organizational activities. Instant messaging represents another communications tool that is rapidly being adopted in organizations. Relying on theories of communication and electronic monitoring, this case study explored the use of IM systems in corporate environments. The study examined why individuals choose IM over other communication media as well as how IM is used in an office environment. Results suggest that critical mass represents an important factor for IM success in the workplace, that IM symbolizes informality, and that IM is perceived to be much less rich than face-to-face communication. Further, results extend past research by demonstrating that employees engage in polychronic communication, view IM as privacy enhancing, and see its interruptive nature as unfair. With the proliferation of instant messaging and other emerging communication technologies in the workplace, this topic represents a rich area for future investigations by organizational psychologists.

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